STUDENT PERFORMANCE ANALYSIS

Using Python, Pandas and NumPy

Student Details	**Information**
Student Name	Adithya K R
Department	Artificial Intelligence and Machine Learning (AIML)
College	JSS Academy of Technical Education, Bengaluru
Course	Python with AI
Project Type	Data Analysis Project

1. Project Title:

Student Performance Analysis Using Pandas and NumPy is a Python-based data analysis project developed to process student academic records and generate meaningful performance insights.

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2. Objective:

The objective of this project is to develop a Student Performance Analysis System using Python, Pandas, and NumPy. The project is created to analyze student academic performance in a simple and organized manner. Instead of manually calculating marks and performance, the system automatically processes student records and generates useful results.

- Store student information in a CSV dataset.
- Load and organize the data using a Pandas DataFrame.
- Calculate total and average marks for each student.
- Assign grades based on average marks.
- Identify whether a student has passed or failed.
- Analyze subject-wise performance.
- Find the highest and lowest-performing students.
- Identify the top-performing students.
- Calculate the overall class average.
- Analyze department-wise performance.
- Consider attendance while determining pass/fail status.

3. Technologies Used:

Technology / Tool	**Purpose**
Python	Main programming language used for implementation
Pandas	Reading CSV files, creating DataFrames and performing data analysis
NumPy	Numerical calculations such as sum, mean, minimum and maximum
CSV	Used to store the student dataset
VS Code	Used for writing and executing the Python program
Git	Used for version control
GitHub	Used to store and share the project repository

The project uses the following Python libraries:

```
import pandas as pd
import numpy as np
```

4. Dataset Description

The project uses a CSV file named students.csv containing 20 student records. The dataset contains student details, marks for different subjects, and attendance percentage.

Column	Description
Student_ID	Unique identification number of the student
Name	Name of the student
Department	Student's department
Python	Marks obtained in Python
Data_Structures	Marks obtained in Data Structures
Database	Marks obtained in Database
Computer_Networks	Marks obtained in Computer Networks
Operating_Systems	Marks obtained in Operating Systems
Attendance	Attendance percentage

The dataset contains students from different departments such as CSE, AIML, ECE, ISE and ME.

Example record:

Student ID	Name	Department	Python	Data Structures	Database	CN	OS	Attendance
S001	Aarav Sharma	CSE	88	91	84	95	92	92%

A total of 20 records are used so that meaningful statistical and performance analysis can be performed.

5. Implementation

Step 1: Prepare the Dataset:

A CSV file named students.csv is created with 20 student records. The file contains student identification details, department, marks in five subjects, and attendance.

Step 2: Load the Dataset

Pandas is used to read the CSV file and store the data in a DataFrame. The program displays the number of records, number of columns, column names, first few records, and basic information.

Step 3: Calculate Total Marks

The marks obtained in all five subjects are added to calculate total marks. NumPy is used for the numerical calculation. The maximum possible total is 500 marks.

Step 4: Calculate Average Marks

The average marks of each student are calculated using NumPy based on marks obtained in all five subjects.

Step 5: Assign Grades

A reusable function assigns grades according to average marks.

Step 6: Identify Pass/Fail

A student is considered Pass when the student scores at least 40 marks in every subject, has an average of at least 50, and has attendance of at least 75%. Otherwise, the student is marked Fail.

Step 7: Subject-Wise Analysis

The project calculates the average, highest and lowest marks for each subject using reusable functions.

Step 8: Identify Top Students

Students are sorted according to average marks and the top three students are displayed.

Step 9: Department-Wise Analysis

Pandas groupby() is used to analyze student performance according to departments.

Step 10: Generate Final Output

The processed results are displayed clearly in the terminal and saved in student_analysis_output.csv.

Example NumPy calculation:

```
marks = df[SUBJECTS].to_numpy(dtype=float)
df["Total_Marks"] = np.sum(marks, axis=1)
```

Grading rule:

Average Marks	Grade
90–100	A+
80–89	A
70–79	B
60–69	C
50–59	D
Below 50	F

Pass condition:

- Minimum 40 marks in every subject.
- Average marks of at least 50.
- Attendance of at least 75%.

6. Key Features

- CSV Dataset Management
- Pandas DataFrame Processing
- NumPy Numerical Calculations
- Total Marks Calculation
- Average Marks Calculation
- Automatic Grade Assignment
- Pass/Fail Identification
- Attendance Analysis
- Highest and Lowest Performance Identification
- Top 3 Student Identification
- Subject-Wise Performance Analysis
- Department-Wise Performance Analysis
- Overall Class Average Calculation
- Reusable Python Functions
- Processed Result Generation
- GitHub Version Control

7. Output / Screenshots

The program produces a clear output containing student performance and analysis results.

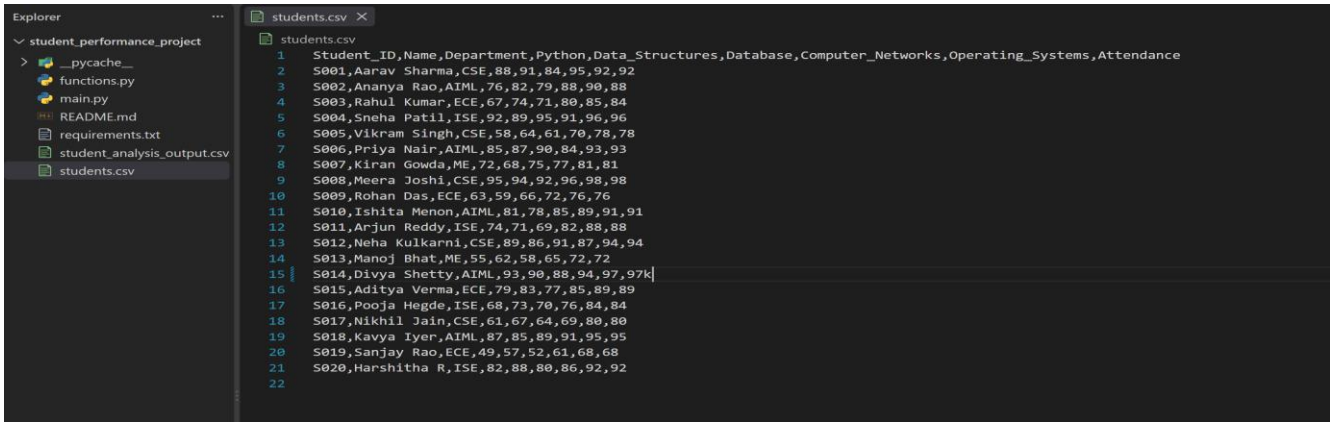
Analysis	**Result**
Total Students	20
Class Average	79.75%
Average Attendance	86.80%
Students Passed	18
Students Failed	2
Highest Average	95.00%
Lowest Average	57.40%
Best Subject	Operating Systems
Best Subject Average	86.95%

Top 3 Students:

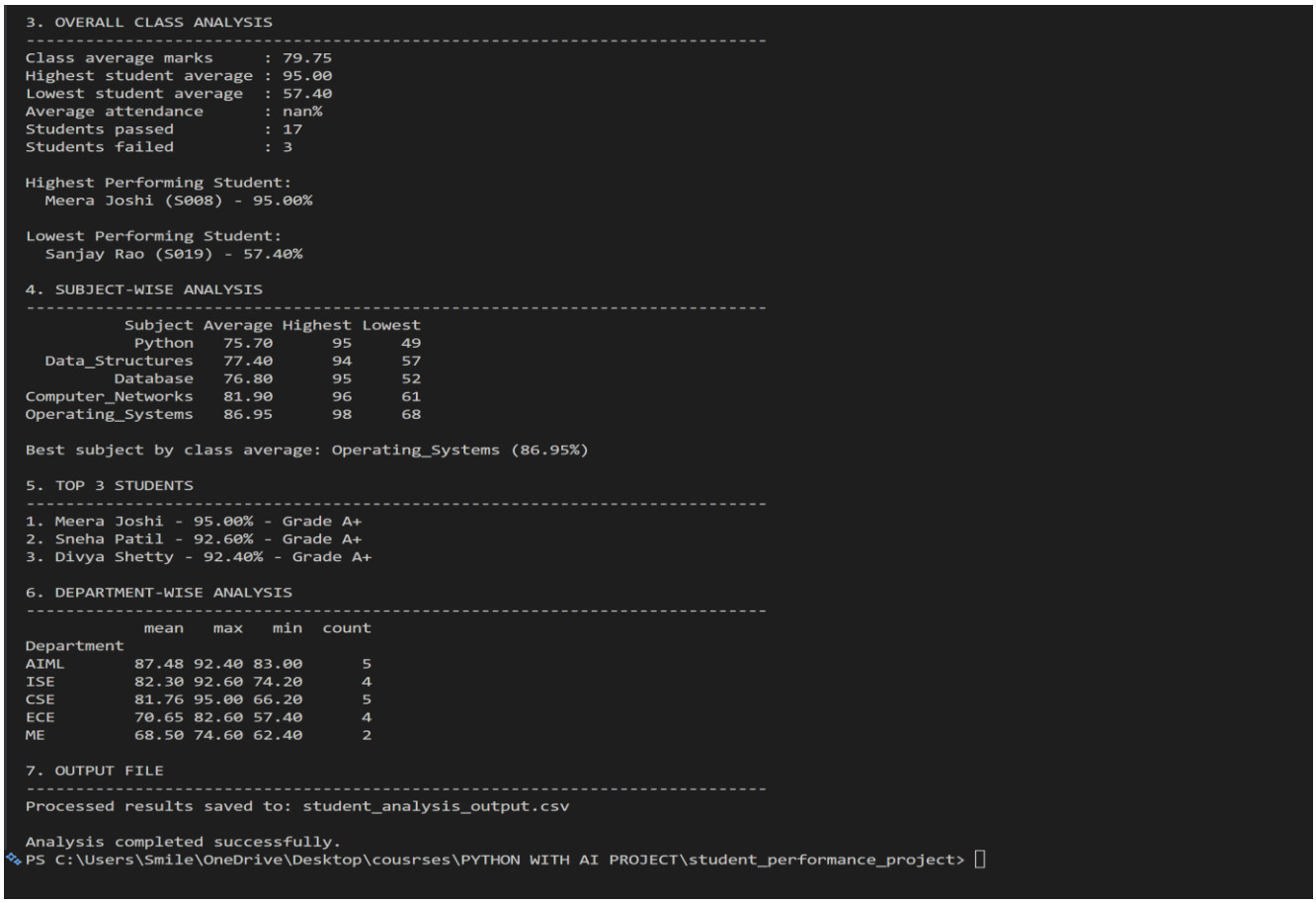
Rank	**Student**	**Average**	**Grade**
1	Meera Joshi	95.00%	A+
2	Sneha Patil	92.60%	A+
3	Divya Shetty	92.40%	A+

Screenshots of some outputs:

- **Screenshot 1: students.csv opened in Excel or VS Code.**



- **Screenshot 2: Student performance output showing total, average, grade and pass/fail, Subject-wise and department-wise analysis**



```
T\s Output (Ctrl+Shift+U) e_project\main.py"
======================================================================
            STUDENT PERFORMANCE ANALYSIS SYSTEM
======================================================================

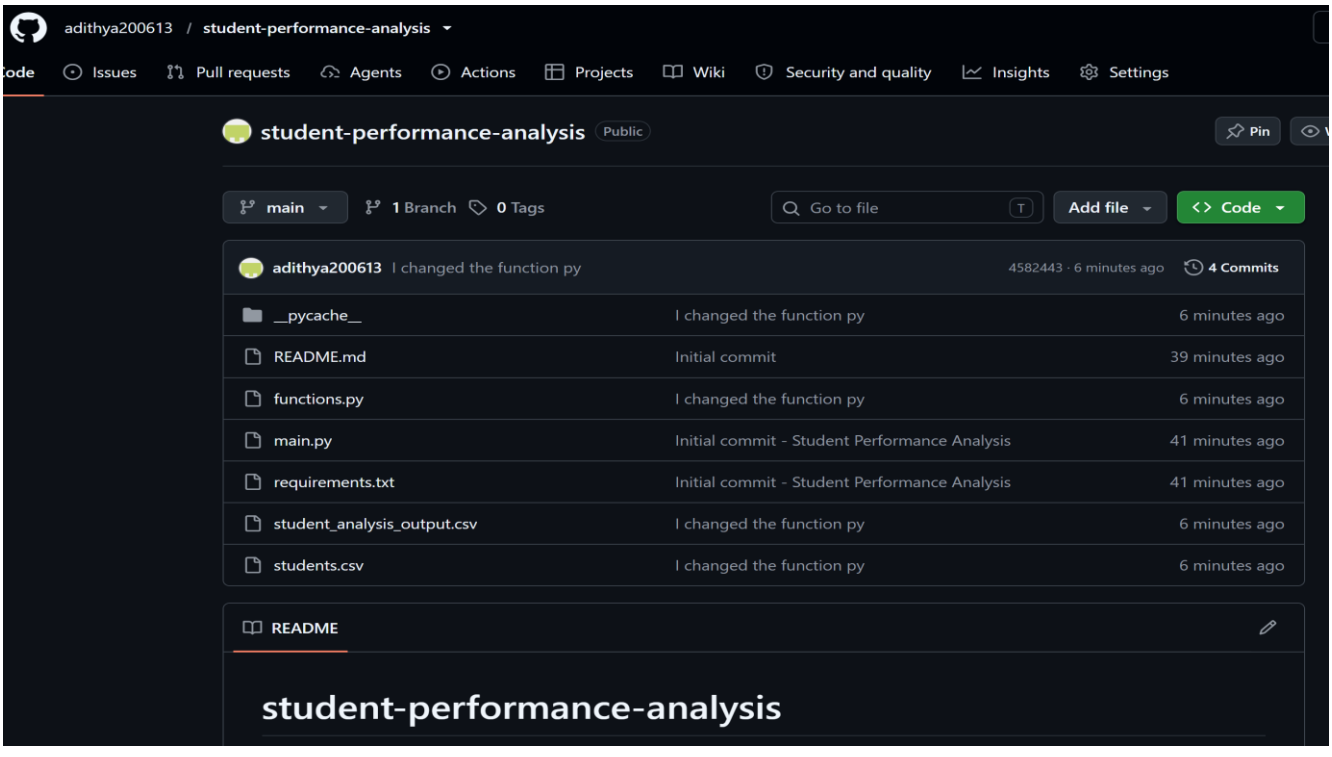
1. DATASET INFORMATION
----------------------------------------------------------------------
Number of records : 20
Number of columns : 9
Columns           : Student_ID, Name, Department, Python, Data_Structures, Database, Computer_Networks, Operating_Systems, Attendance

First 5 Records:
Student_ID         Name Department  Python  Data_Structures  Database  Computer_Networks  Operating_Systems Attendance
      S001 Aarav Sharma        CSE      88               91        84                 95                 92         92
      S002   Ananya Rao       AIML      76               82        79                 88                 90         88
      S003  Rahul Kumar        ECE      67               74        71                 80                 85         84
      S004  Sneha Patil        ISE      92               89        95                 91                 96         96
      S005 Vikram Singh        CSE      58               64        61                 70                 78         78

Basic Information:
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 9 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Student_ID         20 non-null     str
 1   Name               20 non-null     str
 2   Department         20 non-null     str
 3   Python             20 non-null     int64
 4   Data_Structures    20 non-null     int64
 5   Database           20 non-null     int64
 6   Computer_Networks  20 non-null     int64
 7   Operating_Systems  20 non-null     int64
 8   Attendance         20 non-null     str
dtypes: int64(5), str(4)
memory usage: 1.5 KB
None

2. STUDENT PERFORMANCE
----------------------------------------------------------------------
Student_ID         Name Department  Total_Marks Average_Marks Grade  Attendance Pass_Fail
      S001  Aarav Sharma        CSE        450.0         90.00    A+        92.0      Pass
      S002    Ananya Rao       AIML        415.0         83.00     A        88.0      Pass
      S003   Rahul Kumar        ECE        377.0         75.40     B        84.0      Pass
      S004   Sneha Patil        ISE        463.0         92.60    A+        96.0      Pass
      S005  Vikram Singh        CSE        331.0         66.20     C        78.0      Pass
      S006    Priya Nair       AIML        439.0         87.80     A        93.0      Pass
      S007   Kiran Gowda         ME        373.0         74.60     B        81.0      Pass
      S008   Meera Joshi        CSE        475.0         95.00    A+        98.0      Pass
      S009     Rohan Das        ECE        336.0         67.20     C        76.0      Pass
      S010  Ishita Menon       AIML        424.0         84.80     A        91.0      Pass
```

Screenshot 3: GitHub repository showing the project files.



8. Final Outcome

The Student Performance Analysis System successfully analyzes the academic performance of 20 students using Python, Pandas and NumPy.

The project automatically calculates total marks, average marks, grades and pass/fail status. It also performs subject-wise and department-wise analysis and identifies the highest-performing students.

From the analysis, the overall class average was 79.75%, with 18 students passing and 2 students failing. The highest-performing student was Meera Joshi, with an average of 95.00%. Operating Systems was the best-performing subject with an average of 86.95%.

The project demonstrates how Python, Pandas and NumPy can be used to convert raw student data into meaningful information and support quick and accurate academic performance analysis.